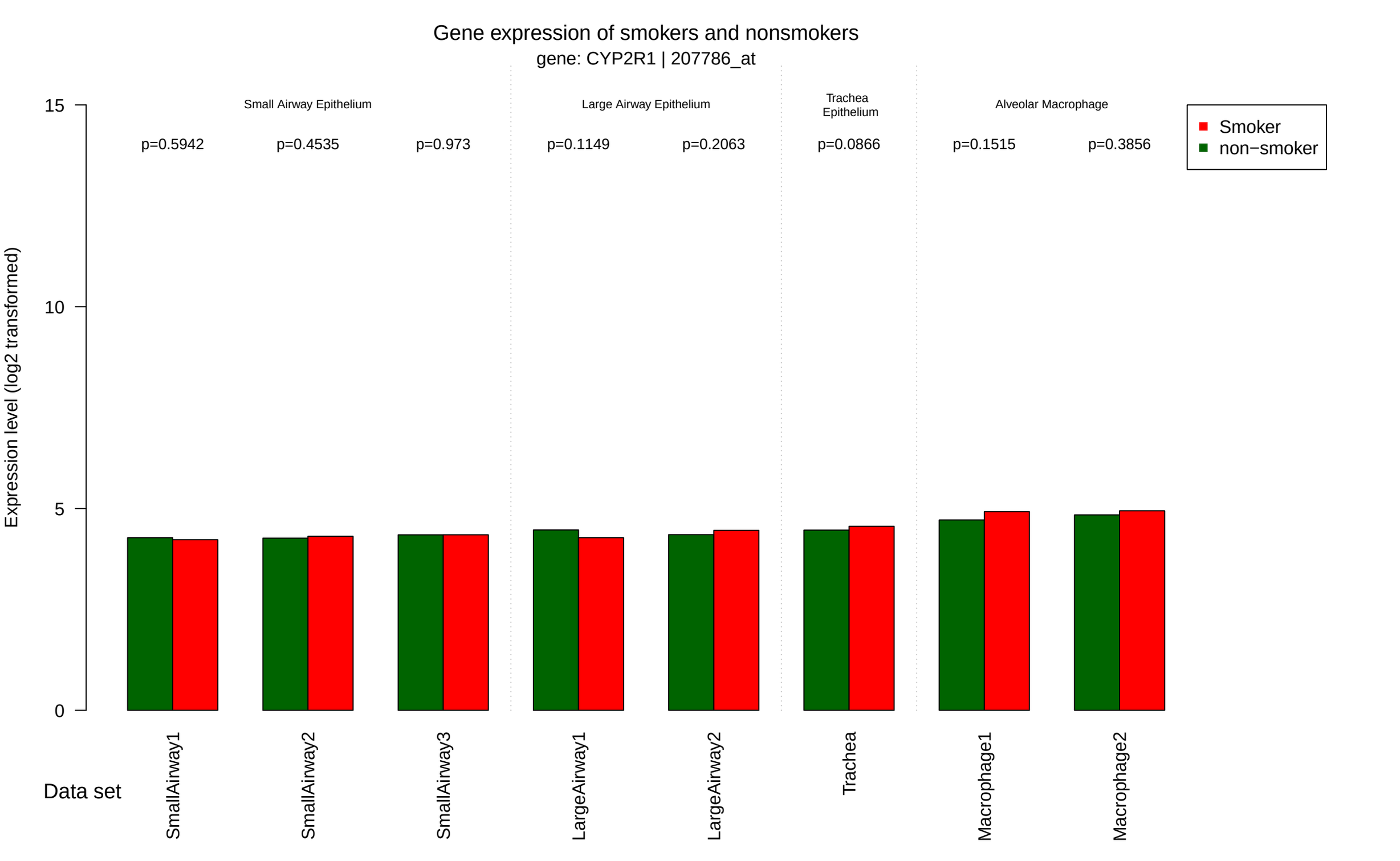
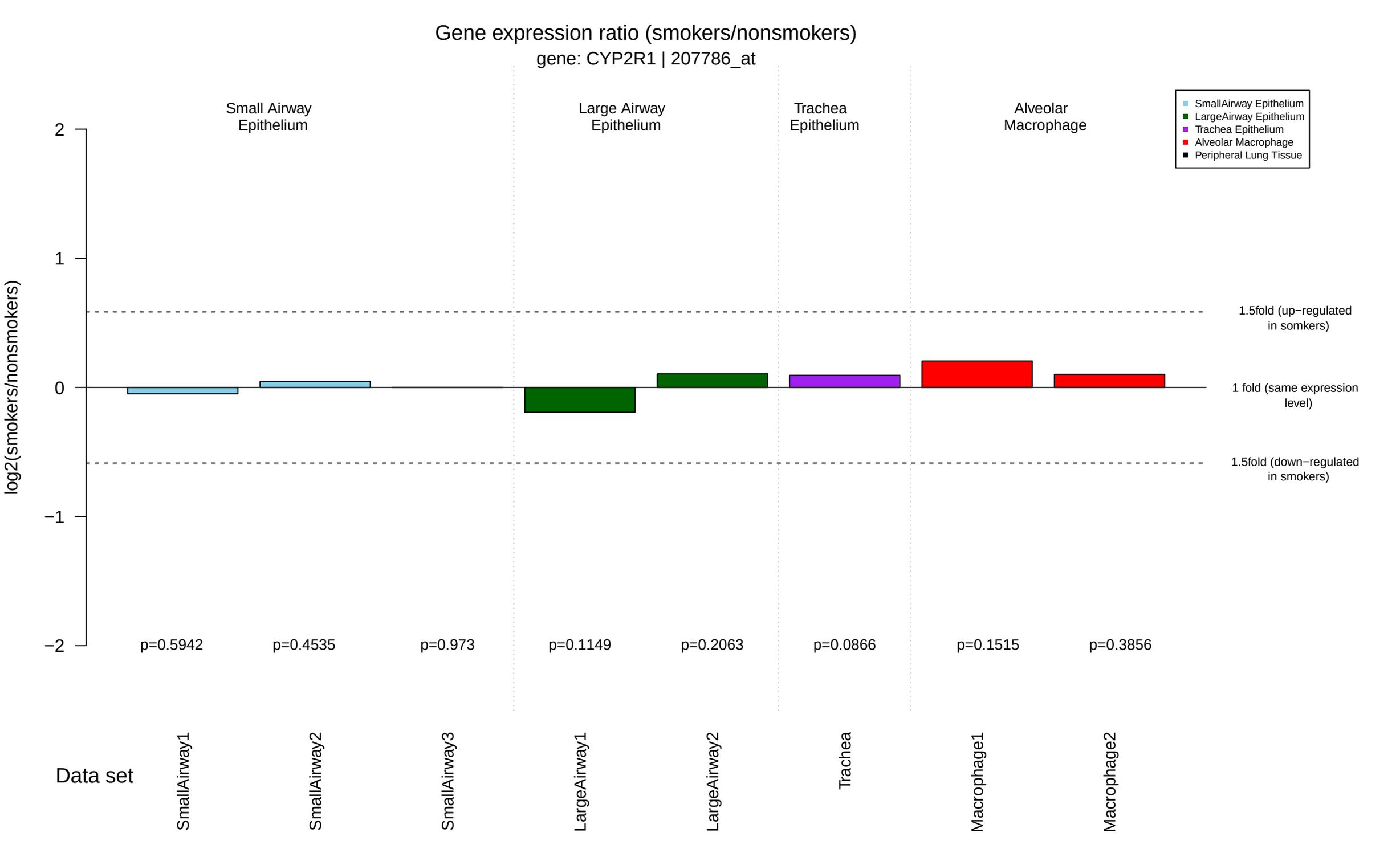
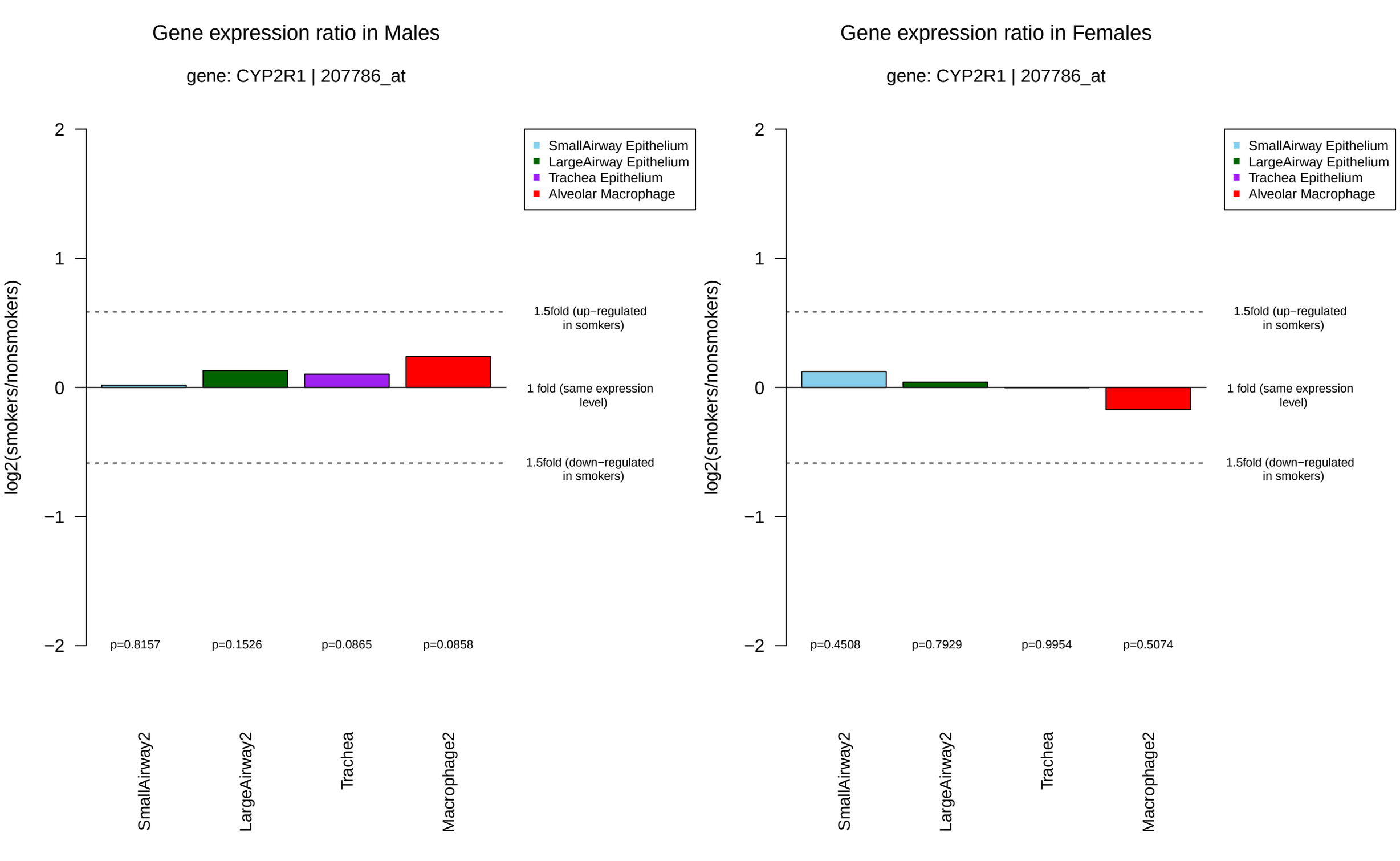
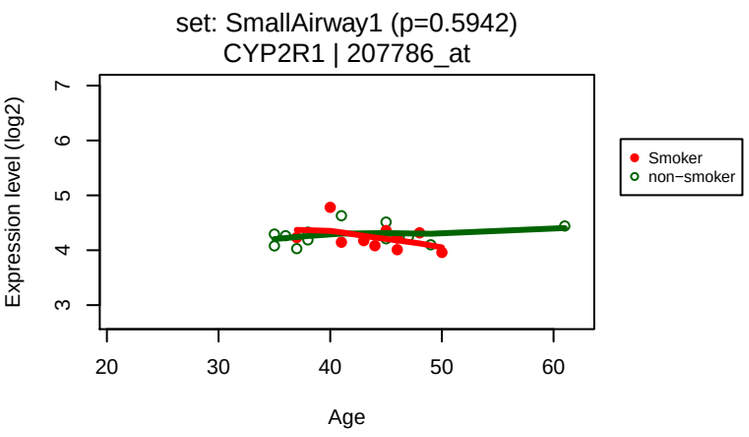
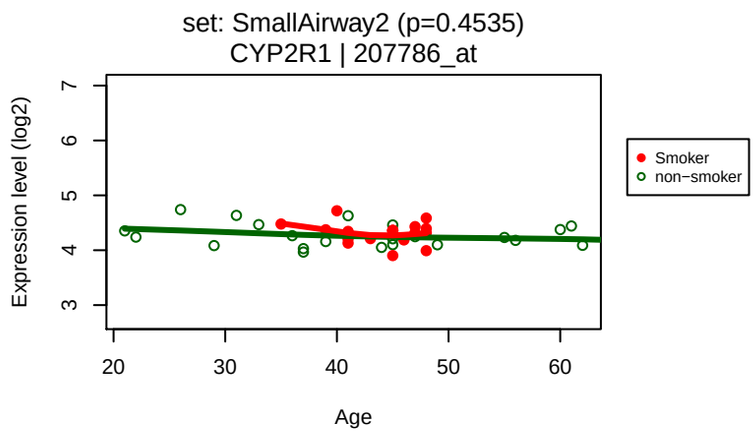
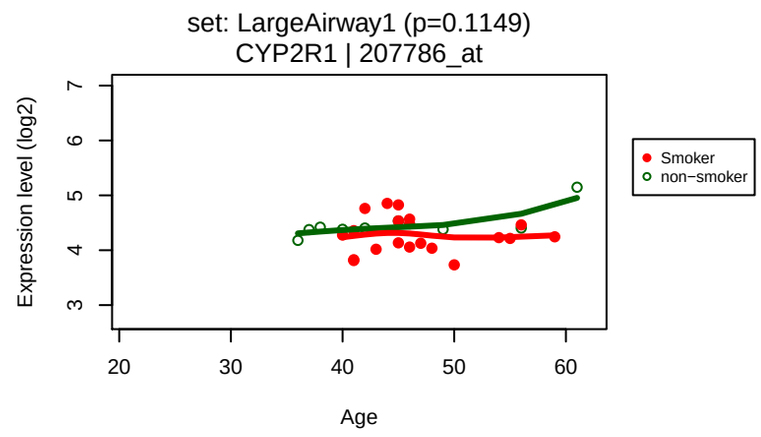

Gene expression of smokers and nonsmokers
gene: CYP2R1 | 207786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5942
p=0.4535
p=0.973
p=0.1149
p=0.2063
p=0.0866
p=0.1515
p=0.3856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

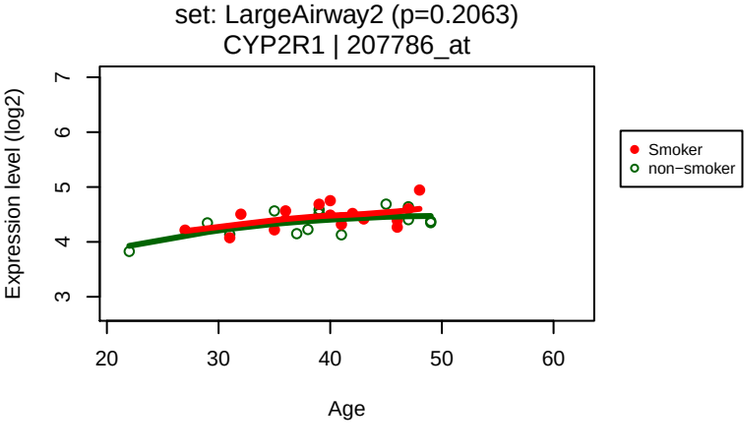

Gene expression ratio (smokers/nonsmokers)
gene: CYP2R1 | 207786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5942
p=0.4535
p=0.973
p=0.1149
p=0.2063
p=0.0866
p=0.1515
p=0.3856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

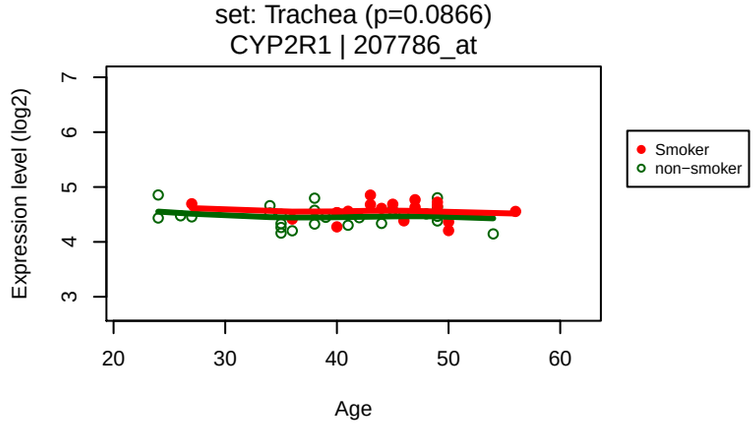

Gene expression ratio in Males
gene: CYP2R1 | 207786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8157
p=0.1526
p=0.0865
p=0.0858
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2R1 | 207786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4508
p=0.7929
p=0.9954
p=0.5074
SmallAirway2
LargeAirway2
Trachea
Macrophage2

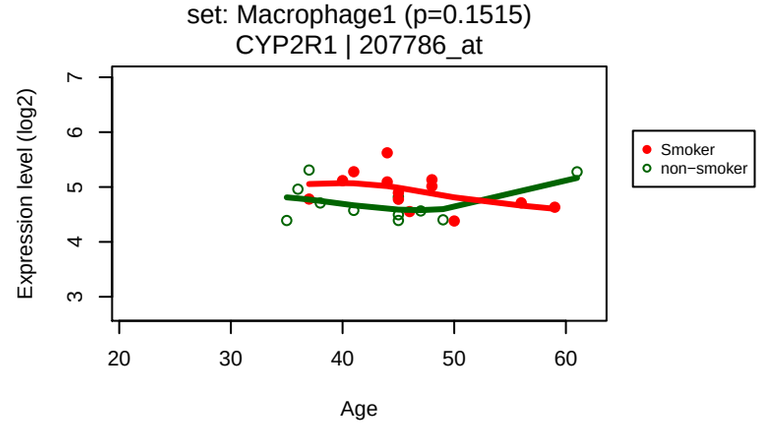

set: SmallAirway1 (p=0.5942)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker

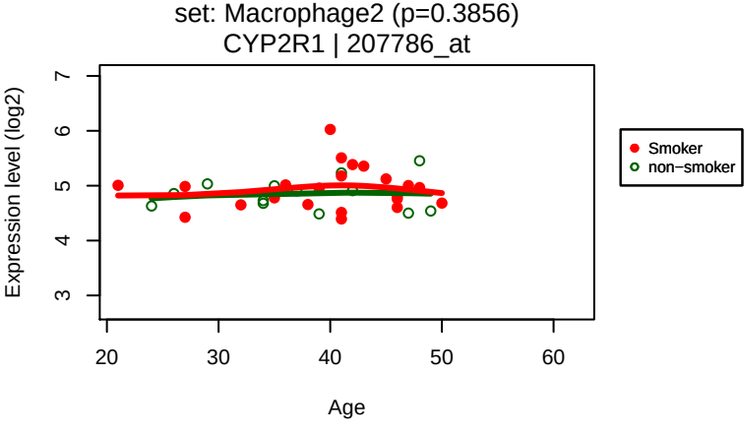

set: SmallAirway2 (p=0.4535)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker

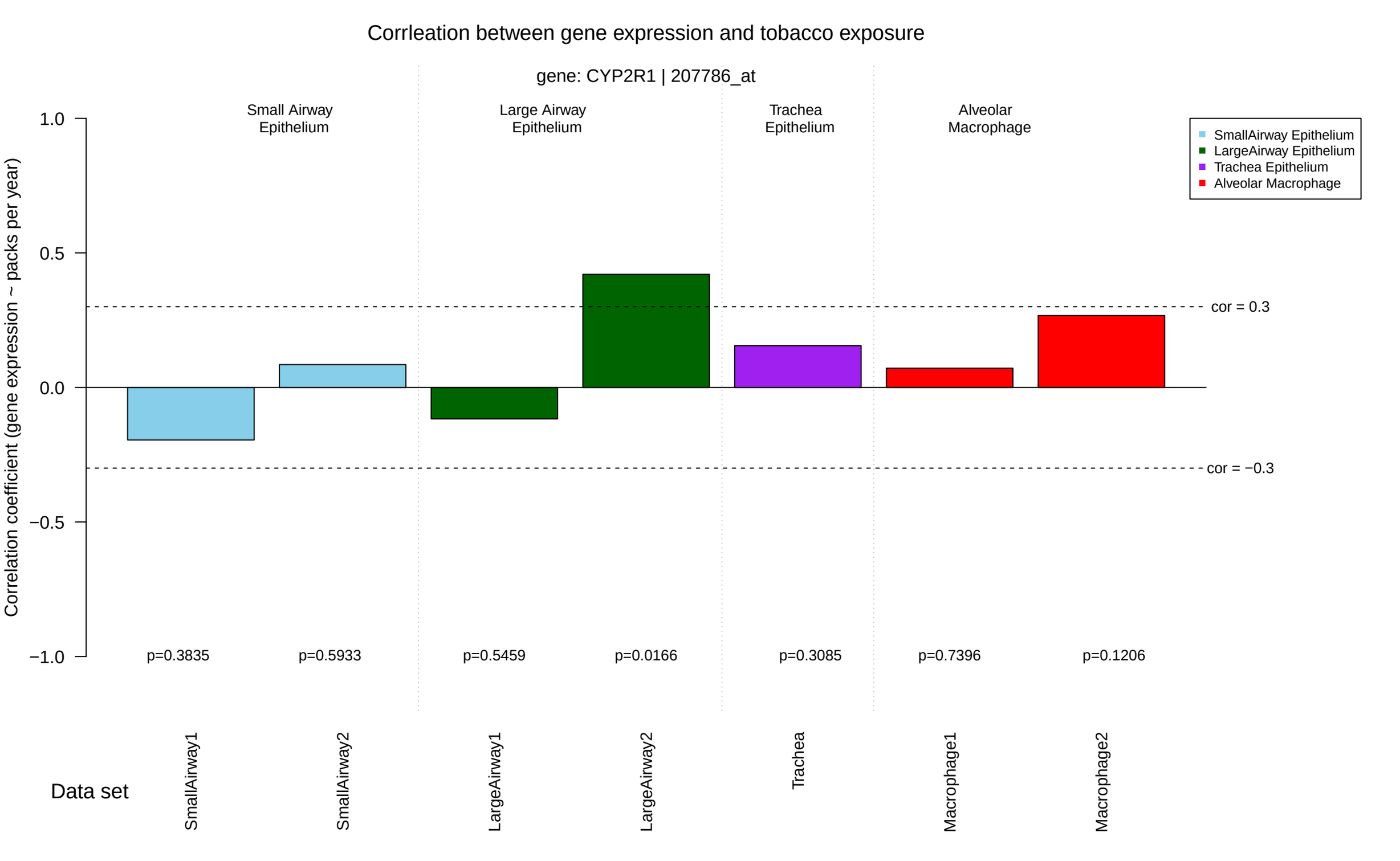

set: LargeAirway1 (p=0.1149)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2063)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0866)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1515)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3856)
CYP2R1 | 207786_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2R1 | 207786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3835
p=0.5933
p=0.5459
p=0.0166
p=0.3085
p=0.7396
p=0.1206
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2